

The Physics of Prospective Learning: Temporal Phase Dynamics and Context Persistence in Biological and Artificial Intelligence

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Abstract

Intelligence has to act before the future is fully observed. Catching a moving object, continuing a plan after interruption, and adapting when a familiar task changes all require more than classifying the present. A system must preserve a future-oriented state, estimate where events are going, compare incoming evidence with that estimate, and revise without either forgetting its objective or becoming rigid.

This paper defines a prospective state as a structured combination of persistent context, fast working state, a trajectory estimate, candidate policies, and mismatch. It then develops a bounded physical hypothesis. Spatial and temporal phase gradients provide measurable direction and propagation variables. Relative phase differences between a projected relation and an observed relation can provide a receiver-specific mismatch signal. Recurrent phase-dependent routing may support rapid revision, while slower structural change can preserve repeatedly useful trajectories.

The paper calls the receiver-relative change a **Phase Wave Differential** only after defining the operation it names. It also recovers four linked ideas from the source history: counterfactual phase configurations, a multi-timescale context stack, contractive mismatch dissipation, and reset amplification. These are connected to established work on predictive coding, active inference, traveling waves, communication through coherence, replay-based planning, fast weights, test-time adaptation, and biological signal amplification. The result is not a universal law of cognition. It is an experimentally discriminable research program whose phase variables must outperform rate, recurrent-state, attention, replay, movement-preparation, and non-phase prediction-error baselines.

Scope and evidence boundary

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1. The problem: acting before the future arrives

Imagine reaching for a cup while someone is moving it. The current retinal image is already old by the time the motor response develops. Successful action requires an estimate of where the cup is going, a representation of the goal, candidate movements, and a way to revise when the trajectory changes.

The same problem appears in an artificial agent. An agent that reconstructs its objective from scratch at every step can drift or repeat itself. An agent that never revises its objective can perseverate after the world changes. The unresolved engineering problem is therefore not merely memory. It is **controlled persistence**: maintain enough state to act prospectively while remaining sensitive to corrective evidence.

Kording and colleagues describe prospective learning as learning for a changing future rather than assuming that future tasks reproduce the training distribution [1]. Existing scientific accounts already supply important parts of the problem. Predictive-coding models compare descending predictions with ascending residual error [2]. The free-energy framework joins prediction, perception, action, and expected trajectories within a

generative model [3]. Prospective coding has also been formalized at the level of spiking neurons [16], and recurrent planning models can use policy rollouts that resemble hippocampal replay [6].

The present paper asks a narrower question that those accounts do not settle: **what measurable physical variables could carry a prospective trajectory, its active context, and the difference between projected and received state?**

Its contribution has five parts:

1. an operational definition of prospective state;
2. a phase-gradient carrier hypothesis for trajectory and routing information;
3. a receiver-relative phase-mismatch variable;
4. a two-timescale account of rapid functional revision and slower consolidation; and
5. failure-capable experiments against simpler explanations.

2. The prospective-state scaffold

Before naming a biological mechanism, define what the system must preserve. Let

$$\mathcal{S}_p(t) = (P(t), W(t), T(t), \Pi(t), F(t)), \quad (1)$$

where:

- $P(t)$ is persistent context or mission state;
- $W(t)$ is fast working or scratch state;
- $T(t)$ is a trajectory estimate;
- $\Pi(t)$ is a set or distribution of candidate policies; and
- $F(t)$ is a mismatch or frustration statistic.

Equation (1) is a measurement scaffold, not a claim that a brain literally stores a five-entry vector. A biological study might operationalize these terms with population activity, relative phase, replay, task rules, behavior, and perturbational responses. An artificial system might use recurrent state, working memory, policy rollouts, and an update gate.

The tuple exposes two opposing failures. If $P(t)$ is too weak, the system repeatedly loses or reconstructs its objective. If $P(t)$ is too rigid, corrective evidence cannot redirect the trajectory. Prospective intelligence requires a controlled exchange between persistent and rapidly revisable state.

3. A physical opportunity: direction in an oscillatory field

Neural activity varies across space and time. Traveling waves are observed at microscopic, mesoscopic, and macroscopic scales, and their phase gradients contain measurable propagation directions [4]. Human connectome topology and connection-strength gradients can shape modeled wave direction and frequency gradients [5]. Human theta and alpha waves also show spatial phase gradients related to task events and performance [13].

Consider a scalar oscillatory field written locally as

$$s(\mathbf{x}, t) = A(\mathbf{x}, t) \cos \phi(\mathbf{x}, t). \quad (2)$$

For a locally planar wave,

$$\phi(\mathbf{x}, t) = \mathbf{k} \cdot \mathbf{x} - \omega t + \phi_0. \quad (3)$$

When $\nabla\phi \neq 0$, the local phase-velocity vector is

$$\begin{aligned} \mathbf{v}_\phi &= -\frac{\partial_t \phi}{\|\nabla\phi\|^2} \nabla\phi; \\ v_\phi &= \frac{\omega}{k} \text{ in one dimension.} \end{aligned} \quad (4)$$

These equations establish a measurable direction and propagation rate. They do not establish that the brain uses phase velocity to represent a future. The prospective hypothesis begins only with the additional claim that a neural receiver may use this direction-bearing variable when maintaining a trajectory, routing an update, or comparing projected activity with incoming activity.

This distinction separates the proposed operation from older vocabulary. A **traveling wave** describes a propagating pattern. **Phase coding** relates information to position within an oscillatory cycle. **Predictive coding** compares predictions and residuals within a hierarchical model. None of those terms by itself specifies the complete receiver-relative chain developed below: projected relation $\rightarrow$ observed relation at a particular receiver $\rightarrow$ measured difference $\rightarrow$ subsequent routing or state change.

4. From expected relation to receiver-specific difference

Suppose a receiving site j has a projected phase relation $\hat{\theta}_j(t)$ and a measured relation $\theta_j(t)$. The wrapped local difference is

$$\Delta\phi_j(t) = \text{wrap}_{(-\pi, \pi]}(\theta_j(t) - \hat{\theta}_j(t)). \quad (5)$$

Across N predefined sites, separate the mean error direction, the dispersion of errors around that direction, and absolute mismatch from the zero-error reference:

$$\begin{aligned} Z_\Delta(t) &= \frac{1}{N} \sum_{j=1}^N e^{i\Delta\phi_j(t)}, \quad \mu_\Delta(t) = \arg Z_\Delta(t), \\ D_\Delta(t) &= 1 - |Z_\Delta(t)|, \quad L_0(t) = \frac{1}{2} \left[1 - \frac{1}{N} \sum_{j=1}^N \cos \Delta\phi_j(t) \right]. \end{aligned} \quad (6)$$

The operation can be stated without specialized vocabulary:

1. a receiver is already in a context-dependent state;
2. prior activity establishes an expected temporal relation;
3. new activity arrives with an observed relation;
4. the receiver is changed by the difference;
5. that change can alter what is routed, suppressed, learned, or projected next.

Self Aware Networks calls this candidate receiver-relative change a **Phase Wave Differential (PWD)**. The term is useful only if the measured difference predicts a consequence beyond simpler variables. Equation (6) deliberately reports different population properties: D_Δ measures error dispersion, while L_0 measures circular mismatch from the projected zero-error relation. The mean direction μ_Δ is undefined when $|Z_\Delta| = 0$. None of these is a universal definition of prediction error or of PWD.

D_Δ is invariant to a common rotation of all errors and therefore cannot by itself measure absolute mismatch. L_0 retains the projected zero-error reference, which must be defined independently of the observations being tested. Both measures can be confounded by amplitude loss, poor phase estimation, volume conduction, filtering, changing unit membership, nonstationarity, or movement. A causal result requires more than correlation: a preregistered mismatch must predict a subsequent update, and perturbing the relevant relation must change that update in the predicted direction.

5. Rapid routing and counterfactual evaluation

Synaptic structure changes more slowly than many acts of attention, task switching, and planning. Relative timing can provide a reversible control layer over a slower structural network. Communication-through-coherence theory proposes that synchronization can selectively alter effective communication between neuronal groups [7]. Human visual-cortex measurements also find predominantly feedforward gamma-band influence and predominantly feedback alpha-beta influence [12].

The prospective extension is not simply that rhythms communicate. It is that a temporary phase configuration may help instantiate one candidate trajectory rather than another. In ordinary language, the system can activate a possible route, compare the route with memory and current evidence, and suppress or strengthen it before overt action. After that operation is understood, the paper calls it **dynamic phase routing** and treats an internally activated candidate as a **counterfactual phase configuration**.

A configuration is not evidence of counterfactual simulation merely because it occurs during planning. It must:

1. appear before action;
2. contain information about a specific candidate future;
3. predict later choice or outcome beyond firing-rate and replay baselines;
4. change appropriately when the candidate policy is experimentally altered; and
5. survive controls for movement preparation, attention, arousal, and sensory expectation.

Recurrent policy-rollout models already explain important features of planning, hippocampal replay, and behavior [6]. They are competitors and implementation alternatives. The phase account earns a role only if phase-resolved measurements or interventions add explanatory power.

6. Context persistence and two-timescale memory

A future-oriented state must remain available across local processing steps. Yet the mechanism that keeps a goal active need not be the mechanism that computes every momentary update. This motivates two functional timescales:

- a **persistent context layer** that changes relatively slowly and constrains the active task;
- a **fast inference layer** that evaluates current evidence, alternatives, and errors.

The source history described the first as a low-frequency or tonic “canvas” and the second as faster phasic “ink.” That metaphor becomes scientifically useful only when each layer has an observable. The slow

component should retain task or trajectory information across interruptions; the fast component should contain update-specific information; and their coupling should predict when the persistent state changes. No frequency band is assigned this role universally in advance.

After the operation is defined, the paper uses **Recurrent State Buffer** as a name for the persistent functional context and **Multi-Frequency Stack** for the proposed interaction between persistent and fast variables.

Longer-term learning adds a second distinction:

Timescale	Proposed function	Candidate measurements	Main alternatives
Fast functional adaptation	Reconfigure active routing without immediate rewiring	Phase relation, effective connectivity, transient state, update gate	Rate change, attention, recurrent dynamics, short-term plasticity
Slow structural consolidation	Stabilize repeatedly useful trajectories	Synaptic efficacy, morphology, replay, settling time, durable behavior	Reward, neuromodulation, ordinary LTP / LTD, replay alone

Fast weights support rapid temporary memory, while test-time training supports rapid parameter adaptation during inference [9,10]. Prioritized replay provides one established way to revisit consequential experience [11]. Prospective configuration provides a further comparison in which neural activity is inferred before weight modification consolidates it [17]. These are not biological identity claims. They show why temporary state, inference-time parameter adaptation, and slower consolidation should be distinguished rather than collapsed into one update type.

Self Aware Networks calls the joined hypothesis **Two-Stage Memory**: rapid phase-dependent adaptation first, slower structural stabilization when a successful pattern recurs. Persistent phase locking alone is not assumed to cause long-term potentiation. The hypothesis requires phase history to predict later stabilization after controlling for firing rate, reward, replay, neuromodulation, and known plasticity variables.

7. Contractive mismatch dynamics and the thermodynamic bridge

The earlier paper used “dissipation” to describe what happens when a discrepancy is repeatedly reduced through local interactions. That idea should not be read as information simply vanishing. The mismatch may be converted into changed state, heat, work, memory, emitted signal, or residual error.

A minimal computational statement is

$$F_{t+1} \leq \rho F_t + \varepsilon_t, \quad 0 \leq \rho < 1, \quad (7)$$

where F_t is a defined mismatch, ρ is a local contraction factor, and ε_t contains new disturbance, model error, or unresolved mismatch. Equation (7) describes successful contractive updating; it does not derive a thermodynamic law.

The thermodynamic bridge becomes an empirical claim only when a study also measures energy or metabolic variables. The paper predicts that successful prospective updating can reduce a specified mismatch while redistributing physical cost. It does not predict that every decrease in phase mismatch lowers total metabolic expenditure, nor that global synchronization is always optimal.

The source history used redness as an illustrative case. In the bounded version, ongoing tonic activity supplies context; sensory input produces phasic departures; recurrent activity changes the population state; and some differences are incorporated into later expectations, action, or memory. This does not imply that redness is

one frequency or one universal phase pattern. The test is whether receiver-relative phase variables explain stimulus-specific state change beyond rate, amplitude, receptive-field, and predictive-coding controls.

The phrase **phase frustration** names a measured phase mismatch only after the measurement model and confounds are specified. “Thermodynamic phase alignment” remains a proposed bridge between contractive computation and physical cost, not a synonym for learning.

8. Reset amplification and biological scaling

A weak distributed signal becomes behaviorally effective only if downstream receivers can detect it. One possible mechanism is a common inhibitory or excitatory event that reduces phase dispersion, allowing aligned components to sum more effectively. Fast-spiking interneuron activation can induce gamma rhythmicity and alter sensory responses [8], but that result does not by itself establish the complete reset mechanism proposed here.

If N coupled units each contribute aligned signal amplitude a , while their noise terms are sufficiently independent with root-mean-square magnitude σ , then

$$\begin{aligned} A_{\text{signal}} &= Na, \\ A_{\text{noise,rms}} &= \sqrt{N} \sigma, \\ \text{SNR} &= \sqrt{N} \frac{a}{\sigma}. \end{aligned} \tag{8}$$

After this conditional operation is stated, the paper calls it **Reset Amplification**. Equation (8) fails when signal phases are not aligned, noise is strongly correlated, the population is not appropriately coupled, or gating selects the wrong units.

Biology clearly can amplify microscopic events: retinal rods produce quantized electrical responses to single-photon absorption [15]. That example establishes molecular-to-cellular signal amplification, not the PWD mechanism. The prospective claim is narrower: after a predefined high-surprise event, an inhibitory or population-reset signature should reduce relevant phase dispersion and improve decoding of the updated trajectory relative to matched non-reset events.

9. Distributed reconstruction and artificial-system analogies

The multiscale aperture synthesis imager reconstructs a coherent optical result from independently operating sensors by computationally estimating and correcting relative phase offsets [14]. This is a useful engineering existence proof for computational phase synchronization across distributed receivers. It is not evidence that neural tissue implements MASI or that optical and neural phase are physically identical.

The analogy clarifies the proposed prospective architecture:

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distributed measurements
-> receiver-specific alignment or mismatch
-> coherent or selectively routed reconstruction
-> updated persistent context
-> action and new measurement
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For artificial agents, the corresponding design requirement is a persistent plan state, a fast scratch state, candidate rollouts, a mismatch variable, and a gate deciding when to revise. “Preserved thinking” is used here as a general design description, not as a claim about any particular commercial model.

10. Experimental program

Test	Primary measurement	Required baselines	Result that weakens the proposal
Phase-gradient trajectory decoding	Out-of-sample prediction of motion or planned trajectory from Equation (4)	Firing-rate, recurrent-state, motion-energy, and non-phase predictive models	Phase variables add no reliable predictive value
Task-switch routing	Phase and cross-frequency change at preregistered switch points	Reaction time, event-related potentials, attention, arousal, and movement covariates	Effects are redundant with ordinary task-switch signals
Projected-versus-observed mismatch	Equations (5)-(6) before and after surprising input	Amplitude, rate, prediction-error, filter, phase-estimation, and membership controls	Mismatch does not predict the subsequent update
Counterfactual state	Pre-action pattern identifying a candidate future	Replay, policy rollout, movement preparation, and random-phase controls	Pattern does not identify or causally influence the candidate future
Persistent-context stack	Slow state retains task information while fast state carries local updates	Recurrent-state, working-memory, band-agnostic, and aperiodic controls	No separable persistence / update roles are found
Contractive update	Defined mismatch follows Equation (7) after corrective evidence	Non-phase error, regression-to-mean, adaptation, and autocorrelation controls	Apparent contraction is not specific to the proposed update
Thermodynamic bridge	Mismatch reduction jointly modeled with metabolic or energy measures	Task difficulty, firing rate, movement, temperature, arousal, and vascular controls	Physical cost does not relate to the preregistered update model
Reset amplification	Dispersion, inhibition, and decoding before / after surprise	Matched non-reset events and correlated-noise models	Reset does not improve trajectory decoding
Phase-structure handoff	Repeated phase pattern followed by later stabilization	Reward, rate, neuromodulation, replay, and plasticity measures	Phase history adds no explanatory value

All analyses should specify recording scale, reference, phase-estimation method, frequency-selection rule, aperiodic-component treatment, multiple-comparison correction, held-out evaluation, and stopping rule. Frequency bands selected after observing the outcome must be labeled exploratory.

11. Relationship to Self Aware Networks and source genealogy

The prospective-learning paper is a focused application of a longer SAN genealogy rather than the origin of every component. The dated source record contains:

- a June 2022 formulation in which neural circuits connect tempo-spatial patterns to predict future inputs and coordinate movement;
- July 2022 formulations of direction, velocity, phase-field trajectories, tonic expectation, and future-oriented memory-prediction;
- a 2022 NAPOT architecture in which distributed arrays receive partial patterns, transform them in their current state, and project phase changes to downstream arrays;
- later syntheses connecting this architecture to prospective learning, multi-timescale state, counterfactual evaluation, contractive mismatch, and reset amplification.

NAPOT is the SAN name for the broader receive-transform-project architecture. PWD names the receiver-relative change within that cycle. A positive prospective-learning result would support one application of

those operators. It would not, by itself, establish consciousness, every NAPOT claim, or the complete SAN architecture.

The companion Claim-Source and Genealogy Atlas records immutable Git commits, source grades, paper-specific claim atoms, inherited scientific background, and later synthesis boundaries.

12. Limitations and interpretation rules

1. The prospective-state tuple is a measurement scaffold, not a uniquely derived state representation.
2. Phase estimates depend on recording scale, reference, filtering, signal-to-noise ratio, and aperiodic activity.
3. Traveling-wave compatibility does not establish prospective representation.
4. Phase effects can be consequences of rate, recurrent dynamics, attention, arousal, sensory expectation, or movement preparation.
5. Error dispersion and zero-referenced circular mismatch are different observables; the projected zero-error relation must be defined independently.
6. The contraction model does not prove a thermodynamic law or guarantee monotonic learning.
7. Reset scaling assumes aligned signal, appropriate coupling, selected receivers, and sufficiently independent noise.
8. Slow and fast roles cannot be assigned to fixed frequency bands without preparation-specific evidence.
9. Optical reconstruction and artificial-system analogies do not establish biological identity.
10. The phase-structure handoff remains a causal hypothesis until phase history adds prediction beyond known plasticity variables.

13. Conclusion

Prospective learning is the physical and computational problem of preserving a useful future-oriented state while remaining correctable. This paper defines that state before naming its candidate mechanisms. It then develops a testable chain:

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persistent context -> projected trajectory -> receiver-relative difference  
-> route or state update -> repeated successful pattern -> slower stabilization  
-> action and new evidence
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Phase gradients provide measurable trajectory variables. PWD names a candidate receiver-relative change. Dynamic phase routing supplies a possible fast control layer. Contractive mismatch dynamics formalize successful correction without claiming that information disappears. Reset amplification gives a conditional population-scaling mechanism. Two-Stage Memory separates rapid functional revision from slower consolidation.

These concepts earn a mechanistic role only if phase-resolved observables and perturbations explain prospective behavior beyond rate, recurrent-state, predictive-coding, replay, attention, and movement baselines. If they do not, the prospective-state problem remains, but this proposed carrier must be narrowed or rejected.

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